A masked-digit lower bound for the Gyarmati–Hennecart–Ruzsa sum–difference constant

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Abstract

The Gyarmati–Hennecart–Ruzsa sum–difference constant quantifies how large a difference set $X-Y$ can be relative to the sumset $X+Y$ for arbitrarily large finite sets X, Y of integers satisfying $|X+Y| \le K|X|$. Zheng's 2025 lower-bound construction encodes bounded integer vectors as base-$(2B+1)$ integers, allows every digit in $\{0, 1, \ldots, B\}$, and imposes a global cap on the sum of the digits. We generalize this construction by allowing the digits to lie in an arbitrary finite mask $M \subseteq \{0, 1, \ldots, B\}$. A standard finite-alphabet multinomial estimate, combined with a carry-free digit argument, reduces the resulting asymptotic lower bound to two finite partition polynomials determined by $M+M$ and, for each $d \in M-M$, by the least possible value of $a+b$ among allowed digits $a, b \in M$ with $a-b=d$. Applying the bound to

$$M = \langle 312, 315, 336, 416, 420\rangle \cap [0, 3084]$$

gives the rigorously certified inequality

$$C_{3a} > 1.19023813.$$

The mask has 901 elements, $|M+M| = 3882$, and $|M-M| = 6003$. It belongs to a structured sequence of numerical-semigroup constructions rather than being an unstructured search output.

1 Introduction and the finite-set principle

Let C_{3a} denote the largest exponent θ for which, for every $K > 1$, there are arbitrarily large finite sets $X, Y \subset \mathbb{Z}$ such that

$$|X+Y| \le K|X|, \qquad |X-Y| \ge c(K)|X+Y|^{\theta}$$

for some $c(K) > 0$. Gyarmati, Hennecart and Ruzsa proved the following finite-set lower-bound principle [3]: if $U \subset \mathbb{N}_0$ is finite and contains zero, then

$$C_{3a} \ge 1 + \frac{\log\big(|U-U|/|U+U|\big)}{\log(2\max U + 1)}. \tag{1}$$

Recent computational work on this constant includes AlphaEvolve [4], Gerbicz's large finite construction [5], and Zheng's large-deviation construction [6]. The *Optimization Constants in Mathematics* repository [2] subsequently listed the finite-certificate bounds 1.1740744 and 1.1835129324 [7, 8]. During preparation of this note, a further open submission to that repository proposed 1.187326127925948 [9]. The bound proved here exceeds all of these values.

Gerbicz introduced bounded base digits into the original construction, and Zheng analyzed the limiting exponential growth of a bounded-digit family. In Zheng's version the allowed digit alphabet is the complete interval $\{0,\dots,B\}$. We replace that interval by a mask M. This can remove many possible sum digits while preserving difference digits d that admit witnesses $a,b\in M$ with $a-b=d$ and small $a+b$, so that they use little of the two digit-sum budgets in the construction. The multinomial counting estimate used below is standard; Proposition 1 is the construction-specific arbitrary-mask consequence.

2 The masked-digit bound

Let $M\subset\mathbb{N}_0$ be finite with $0\in M$ and $B=\max M\geq 1$. Let

$$Q=2B+1,\qquad S=M+M,\qquad D=M-M.$$

For each $d\in D$, define its minimum witness cost

$$\kappa(d)=\min\{a+b:a,b\in M,\ a-b=d\}.\tag{2}$$

The function κ is even: $\kappa(-d)=\kappa(d)$. Define

$$P_+(x)=\sum_{s\in S}x^s,\qquad P_-(x)=\sum_{d\in D}x^{\kappa(d)}.\tag{3}$$

We use the following standard lower bound for a multinomial coefficient; see, for example, Csiszár [1, Lemmas II.1 and II.2]. If E is a nonempty finite set and nonnegative integers r_e satisfy $\sum_{e\in E}r_e=m$, let $q_e=r_e/m$. Then

$$\log\frac{m!}{\prod_{e\in E}r_e!}\geq -m\sum_{e\in E}q_e\log q_e-\log\binom{m+|E|-1}{|E|-1},\tag{4}$$

where log is the natural logarithm and $0\log 0$ is interpreted as zero. For fixed E, the binomial coefficient in (4) is a polynomial in m of degree $|E|-1$; equivalently,

$$\log\binom{m+|E|-1}{|E|-1}=O_E(\log m)=o(m).$$

Proposition 1 (Masked-digit bound)**.** *For every $x\in(0,1)$,*

$$C_{3a}\geq 1+\frac{\log P_-(x)-\log P_+(x)}{\log(2B+1)}.\tag{5}$$

Consequently, one may take the supremum of the right-hand side over $x\in(0,1)$.

Proof. Fix $x\in(0,1)$ and define

$$p_d=\frac{x^{\kappa(d)}}{P_-(x)}\qquad(d\in D).\tag{6}$$

Since κ is even, these weights satisfy $p_d=p_{-d}$. Let

$$\mu=\sum_{d\in D}p_d\kappa(d).$$

For each $d\in D$, choose $\alpha_d,\beta_d\in M$ such that

$$\alpha_d-\beta_d=d,\qquad\alpha_d+\beta_d=\kappa(d),$$

with $(\alpha_{-d}, \beta_{-d}) = (\beta_d, \alpha_d)$; for $d = 0$, take $(\alpha_0, \beta_0) = (0, 0)$.

Let

$$R = \log \frac{P_-(x)}{P_+(x)}.$$

If $R \le 0$, then (5) follows from $C_{3a} \ge 1$, which is obtained from (1) using $U = \{0, 1\}$. Assume henceforth that $R > 0$, and choose

$$0 < \varepsilon < \frac{R}{-\log x}.$$

For each positive $d \in D$, define

$$n_d^{(m)} = n_{-d}^{(m)} = \lfloor m p_d \rfloor, \qquad n_0^{(m)} = m - 2 \sum_{\substack{d \in D \\ d > 0}} \lfloor m p_d \rfloor.$$

These are nonnegative integers, their sum is m, and $n_d^{(m)}/m \to p_d$ for every $d \in D$. Write $q_d^{(m)} = n_d^{(m)}/m$. The number of sequences containing exactly $n_d^{(m)}$ copies of each d is the multinomial coefficient

$$\frac{m!}{\prod_{d \in D} n_d^{(m)}!}.$$

By (4), its logarithm is at least

$$-m \sum_{d \in D} q_d^{(m)} \log q_d^{(m)} - \log \binom{m + |D| - 1}{|D| - 1} = mG(p) + o(m), \qquad G(p) = -\sum_{d \in D} p_d \log p_d.$$

Indeed, $q_d^{(m)} \to p_d$ for every d, the function $t \mapsto -t \log t$ is continuous on $[0, 1]$ when its value at zero is taken to be zero, and the final logarithmic term is $o(m)$ because D is fixed. Also, for all sufficiently large m,

$$\sum_{d \in D} n_d^{(m)} \kappa(d) \le (\mu + \varepsilon) m.$$

Set

$$L_m = \left\lfloor \frac{(\mu + \varepsilon) m}{2} \right\rfloor$$

and define

$$U_m = \left\{ \sum_{i=0}^{m-1} a_i Q^i : a_i \in M, \ \sum_{i=0}^{m-1} a_i \le L_m \right\}. \tag{7}$$

Take any sequence $(d_0, \dots, d_{m-1})$ containing exactly $n_d^{(m)}$ copies of each $d \in D$. Symmetry of the counts gives $\sum_i d_i = 0$. Hence

$$\sum_i \alpha_{d_i} - \sum_i \beta_{d_i} = 0, \qquad \sum_i \alpha_{d_i} + \sum_i \beta_{d_i} = \sum_i \kappa(d_i),$$

so each of the first two sums equals one half of the last. Each is an integer not exceeding $(\mu + \varepsilon) m/2$, and hence is at most L_m. Therefore

$$u = \sum_{i=0}^{m-1} \alpha_{d_i} Q^i, \qquad v = \sum_{i=0}^{m-1} \beta_{d_i} Q^i$$

belong to U_m, and

$$u - v = \sum_{i=0}^{m-1} d_i Q^i.$$

The map from difference-digit sequences to integers is injective. Indeed, if

$$\sum_i d_i Q^i = \sum_i d_i' Q^i,$$

then, with $\delta_i = d_i - d_i'$, we have $\sum_i \delta_i Q^i = 0$ and $|\delta_i| \le 2B < Q$. Reduction modulo $Q = 2B+1$ therefore gives $\delta_0 = 0$; after division by Q, induction gives $\delta_i = 0$ for every i. The multinomial estimate above therefore yields

$$\liminf_{m\to\infty} \frac{1}{m} \log|U_m - U_m| \ge G(p). \tag{8}$$

From (6),

$$G(p) = \log P_-(x) - \mu \log x. \tag{9}$$

Every element of $U_m + U_m$ has a carry-free base-Q digit sequence $(s_0, \dots, s_{m-1}) \in S^m$ with $\sum_i s_i \le 2L_m$. If N_m is the number of all such sequences, then, because $0 < x < 1$,

$$N_m x^{2L_m} \le \sum_{(s_i)\in S^m} x^{\sum_i s_i} = P_+(x)^m.$$

Thus

$$\limsup_{m\to\infty} \frac{1}{m} \log|U_m + U_m| \le \log P_+(x) - (\mu + \varepsilon) \log x. \tag{10}$$

Combining (8), (9), and (10) gives

$$\liminf_{m\to\infty} \frac{1}{m} \log \frac{|U_m - U_m|}{|U_m + U_m|} \ge R + \varepsilon \log x > 0. \tag{11}$$

Finally, every $u \in U_m$ has digits at most B, so

$$u \le B \sum_{i=0}^{m-1} Q^i = \frac{Q^m - 1}{2}.$$

Consequently,

$$2 \max U_m + 1 \le Q^m.$$

The numerator in (1) is positive for all sufficiently large m by (11); hence replacing its denominator by the larger quantity $m \log Q$ preserves a lower bound. Applying (1) and taking the lower limit of the resulting right-hand side gives

$$C_{3a} \ge 1 + \frac{R + \varepsilon \log x}{\log Q}.$$

Letting $\varepsilon \downarrow 0$ proves (5). □

Remark 1 (Recovery of Zheng's optimization)**.** *For the complete mask $M = \{0, 1, \dots, B\}$,*

$$P_+(x) = 1 + x + \cdots + x^{2B}, \qquad P_-(x) = 1 + 2x + 2x^2 + \cdots + 2x^B.$$

Maximizing Proposition 1 for $B = 5$ reproduces Zheng's value 1.173077279785… *[6]. Thus the proposition is a direct mask generalization of Zheng's limiting construction, although the proof above is self-contained apart from the finite-set principle* (1) *and the standard multinomial estimate* (4).

3 The certified mask

Let

$$H = \langle 312, 315, 336, 416, 420\rangle = \{312a + 315b + 336c + 416d + 420e : a, b, c, d, e \in \mathbb{N}_0\} \tag{12}$$

and let

$$M = H \cap [0, 3084].$$

Exact integer calculation gives

$$B = \max M = 3084, \qquad Q = 2B + 1 = 6169,$$

and

$$|M| = 901, \qquad |M + M| = 3882, \qquad |M - M| = 6003. \tag{13}$$

The conductor of H is 4574, and the cheapest witness for difference one has cost

$$\kappa(1) = 1463.$$

The verification package contains the complete mask, complete sum support, and all values $(d, \kappa(d))$ for $d \in M - M$.

At the fixed decimal value

$$\lambda_0 = 0.0016039887760343438, \qquad x_0 = e^{-\lambda_0},$$

the partition sums are approximately

$$P_-(x_0) = 229.6998248684831596707\ldots, \qquad P_+(x_0) = 43.66337113593047602635\ldots,$$

so

$$\log \frac{P_-(x_0)}{P_+(x_0)} = 1.66026378574340855999\ldots$$

and

$$1 + \frac{\log(P_-(x_0)/P_+(x_0))}{\log 6169} = 1.19023813803716219098\ldots. \tag{14}$$

The accompanying MPFR verifier uses 384-bit arithmetic with directed rounding. It reconstructs all discrete data from the five generators and proves the conservative strict inequality

$$\boxed{C_{3a} > 1.19023813}. \tag{15}$$

No optimization claim is needed for the proof: Proposition 1 holds at the single fixed value λ_0.

4 Structure of the successful masks

A numerical semigroup is a cofinite additive submonoid of $\mathbb{N}_0$. The record mask emerged from a sequence of numerical-semigroup masks with increasingly rich but related structure.

4.1 First structured mask

The first strong mask was

$$H_1 = 12\langle 2, 3\rangle + 13\langle 2, 3\rangle = \langle 24, 26, 36, 39\rangle, \qquad B = 189,$$

which gives

$$C_{3a} \geq 1.1893936243\ldots.$$

4.2 Four-generator improvement

The next improvement was

$$H_2 = 20\langle 3,4\rangle + 21\langle 3,4\rangle = \langle 60,63,80,84\rangle, \qquad B = 538,$$

which gives

$$C_{3a} \geq 1.1902074763\ldots.$$

For the general family $S\langle k,k+1\rangle + (S+1)\langle k,k+1\rangle$, the generators satisfy

$$Sk + (S+1)(k+1) = S(k+1) + (S+1)k + 1.$$

Thus two allowed sums differ by one, helping to keep the difference alphabet dense while the semigroup gaps suppress the sum alphabet.

4.3 Five-generator record and iterated gluing

For numerical semigroups Γ_1, Γ_2 and coprime positive integers d_1, d_2 with $d_1 \in \Gamma_2$ and $d_2 \in \Gamma_1$, the semigroup $d_1\Gamma_1 + d_2\Gamma_2$ is called a gluing. This operation is standard in the theory of complete-intersection numerical semigroups [10].

Let

$$\Gamma = \langle 3,4\rangle, \qquad A = \langle 15,16,20\rangle.$$

First,

$$A = 5\Gamma + 16\mathbb{N}_0.$$

This is a gluing of Γ and $\mathbb{N}_0 = \langle 1\rangle$: the coefficient 5 belongs to $\mathbb{N}_0$, the coefficient 16 belongs to Γ ($16 = 4 \cdot 4$), and $\gcd(5,16) = 1$. Second,

$$H = 21A + 104\Gamma.$$

This is a gluing of A and Γ: the coefficient 21 belongs to Γ ($21 = 7 \cdot 3$), the coefficient 104 belongs to A ($104 = 4 \cdot 16 + 2 \cdot 20$), and $\gcd(21,104) = 1$. Thus the displayed decomposition exhibits H as an iterated gluing. Since $\mathbb{N}_0$ and $\langle 3,4\rangle$ are complete intersections and gluing preserves the complete-intersection property, H is a complete-intersection numerical semigroup [10].

Four of the five generators form a larger copy of the same unit-defect core:

$$\langle 312,315,416,420\rangle = 104\langle 3,4\rangle + 105\langle 3,4\rangle,$$

and

$$312 + 420 = 315 + 416 + 1.$$

The additional generator $336 = 21 \cdot 16$ selectively changes the gap and bridge structure. This nested pattern suggests that the present result may belong to a broader family. Searches of several natural nearby families have not exhausted all structured gluings, higher embedding dimensions, or multilevel digit constraints.

5 Verification and reproducibility

The source code, certificate files, and verification scripts accompany this paper. Before public release they will be archived at a permanent location, whose URL will replace the following placeholder:

`PACKAGE_URL_TO_BE_ADDED`.

The package contains the following independently written checks.

1. `verification/verify_mpfr.cpp` reconstructs M, $M + M$, and $\kappa(d)$ using exact integer arithmetic and certifies (15) using MPFR directed rounding.
2. `verification/verify_python.py` independently reconstructs the exact discrete data using the Python standard library and checks the shipped certificate files and hashes.
3. `verification/verify_gp.gp` independently reconstructs the data in PARI/GP and evaluates the bound at high precision. This is a secondary numerical cross-check; the MPFR program supplies the rigorous floating-point certificate.

Acknowledgements and AI-use disclosure

During this work, the author used several large language models (LLMs), as follows: OpenAI's GPT-5.6 Sol model was used extensively in software development, analysis of search results, writing this paper, and preparation of the verification package; Google's Gemini 3.1 Pro Preview was used in software development and analysis of search results; Google's Gemini 3.5 Flash was used to review this paper; Anthropic's Sonnet 5 was used in analysis of search results. The author guided the software development and analysis of search results, noticed emerging structure in promising masks, and prompted the LLMs to reason about generalizations that could yield a stronger result; only GPT-5.6 Sol successfully generalized in a way that achieved a stronger result. The author ran the search and verificaton software, reviewed this paper, and takes responsibility for the final mathematical claims and references.

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